

ocean Ai

// session 01 beginner Python

Setup & variables

Name your ship. Change the journey. Save your first version.

VS Code

Python

GitHub

// welcome aboard



Leave with something **real**.

By the end of today, you will have:

01

Set up

Install the same tools used by real software teams.

02

Run

Launch a working Python game from the terminal.

03

Change

Personalise the ship using variables.

04

Save

Push your first commit to GitHub.

OUTCOME

Your game. Your first commit.

One game. Nine sessions.

You command a ship, cross a galaxy,
manage resources, and try to reach the final
destination alive.

9

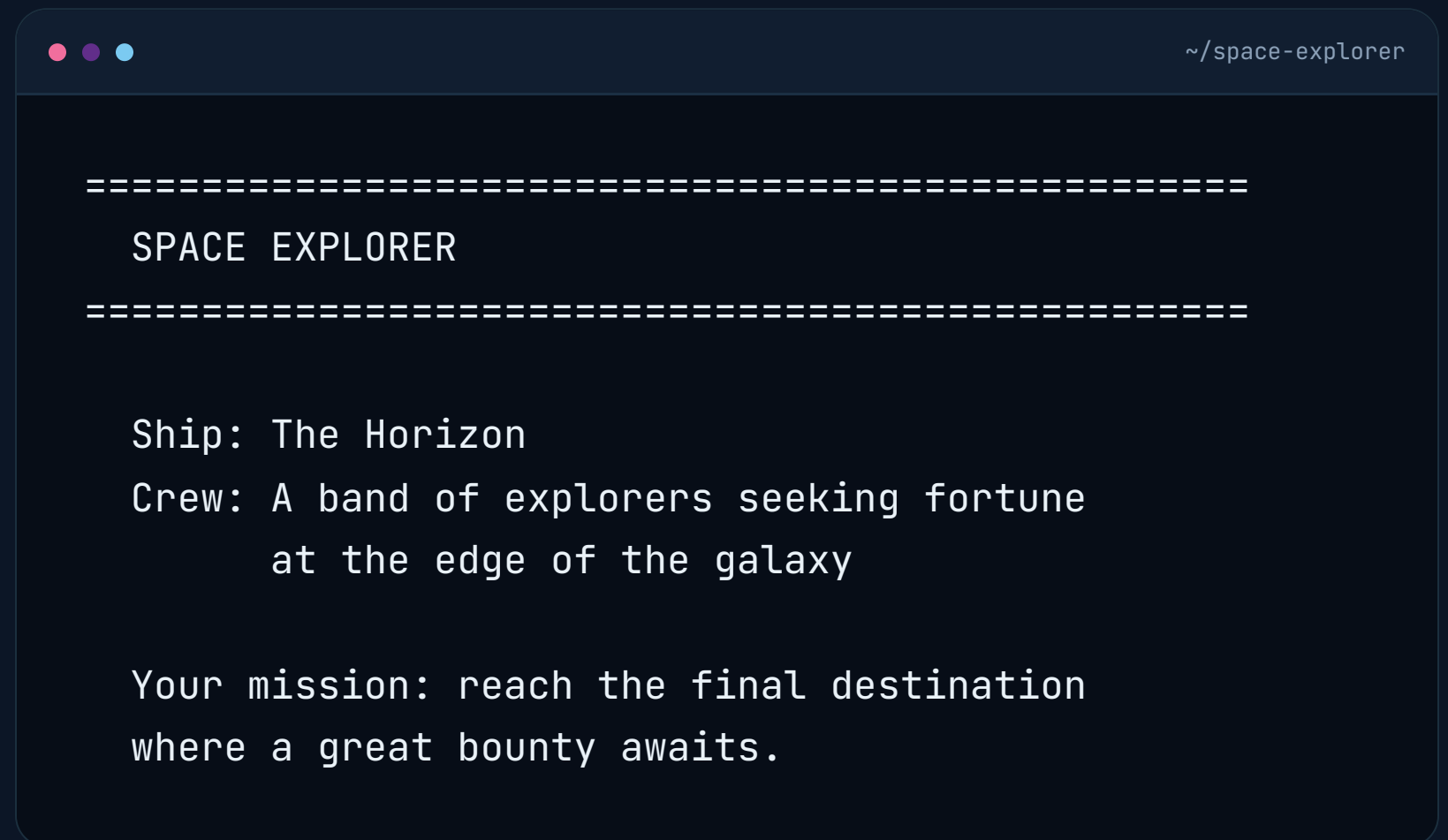
sessions

1

project

0→1

learning path



The engine fades out. Your code grows in.



TODAY Change the information the game starts with.



01

// watch first then build

What do you notice?

```
$ uv run game.py
```

ship

destinations

oxygen

hull

No typing yet. Let the engine fly.

Our workshop

Three tools, each with one clear job.



VS Code

Where we read and write code.

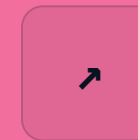
editor



uv

Runs Python and manages the environment.

runner



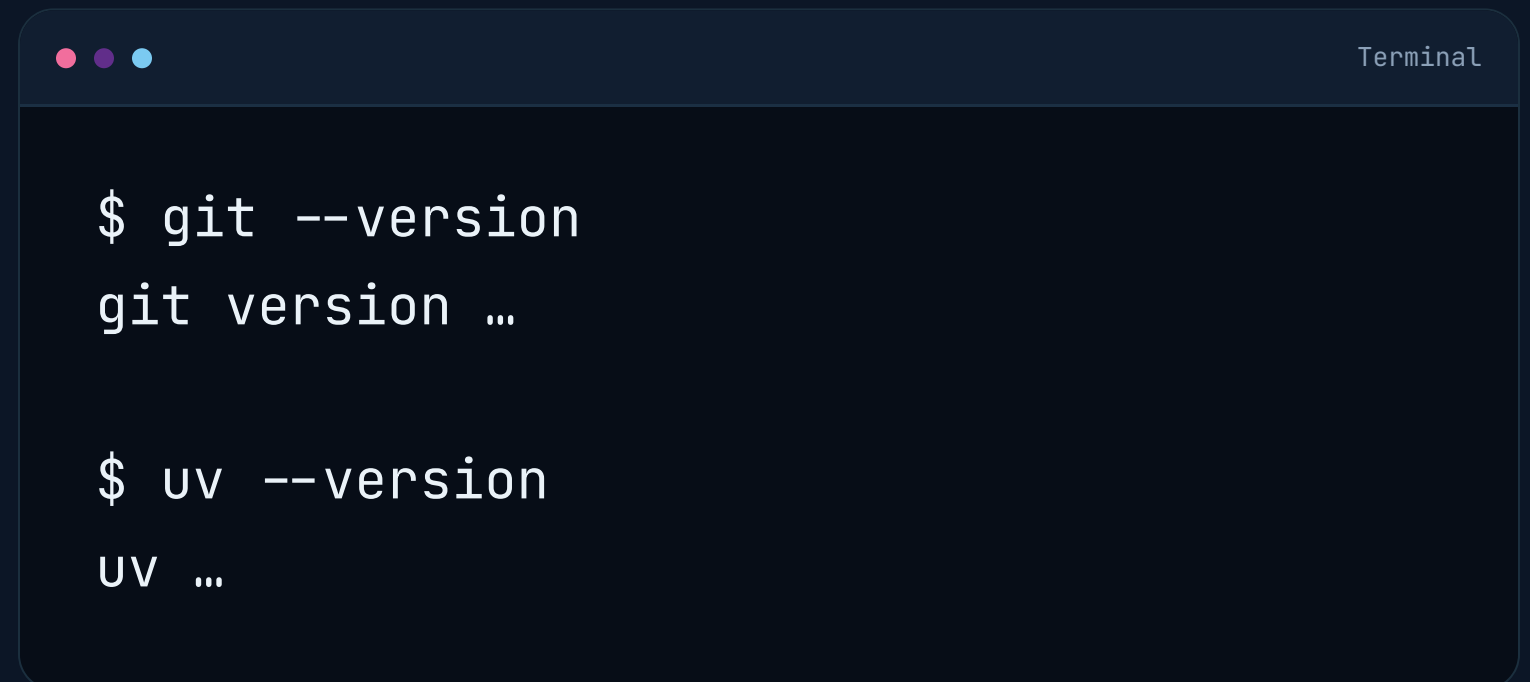
Git + GitHub

Tracks our progress and shares it.

history

Ask each tool: “Are you ready?”

Your version numbers may be different. A version is success; an error needs help.

A terminal window with a dark blue background and a title bar labeled "Terminal". The window contains two command-line sessions. The first session shows the command "\$ git --version" followed by the output "git version ...". The second session shows the command "\$ uv --version" followed by the output "uv ...".

```
$ git --version
git version ...

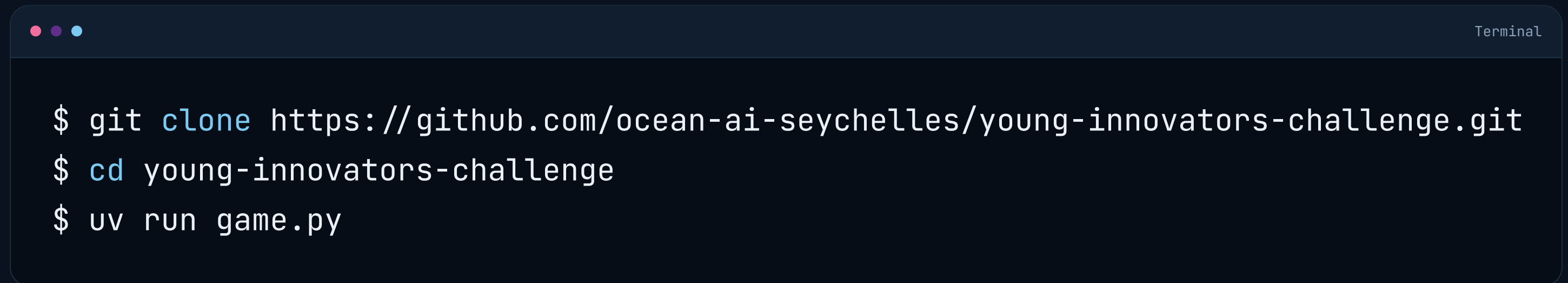
$ uv --version
uv ...
```

CHECKPOINT Both commands print a version.

// terminal get the game

Clone. Enter. Launch.

Use the repository address provided by your instructor.



```
$ git clone https://github.com/ocean-ai-seychelles/young-innovators-challenge.git
$ cd young-innovators-challenge
$ uv run game.py
```

`clone` copy it

`cd` enter it

`run` launch it



OK

```
// checkpoint the game runs
```

You just ran an unfamiliar Python project.

That is real development work.

SPACE EXPLORER appears in the terminal

Five values control today's mission.

We will change the information—not the engine.

constants.py

```
SHIP_NAME = "The Horizon"  
CREW_DESCRIPTION = "A band of explorers seeking  
fortune at the edge of the galaxy"  
  
STARTING_OXYGEN = 100  
STARTING_HULL = 100  
  
GALAXY_SIZE = 8
```

A variable is a **name** that stores a **value**.

SHIP_NAME

=

"The Horizon"

name

what the game knows

store

put this
value here

value

the information

Python notices what kind of value you store.

```
ship_name = "The Voyager"      # string
starting_oxygen = 100          # int
speed = 2.5                    # float
```

string	text in quotes	"The Voyager"
---------------	----------------	---------------

int	a whole number	100
------------	----------------	-----

float	a decimal number	2.5
--------------	------------------	-----



```
// predict before we run
```

What happens if the quotes disappear?

```
SHIP_NAME = "The Voyager"
```

→

```
SHIP_NAME = The Voyager
```

Python stops reading it as text.
Keep opening and closing quotes around strings.

STEP 01

Name the ship.

Edit only the value on the right. Keep both quotes.

```
SHIP_NAME = "The Voyager"
```

```
$ uv run game.py
```

Look for **Ship: The Voyager** in the intro.

STEP 02

Shorten the journey.

Use a positive whole number. Save, run, then count the destinations.

```
GALAXY_SIZE = 5
```

EXPECTED

[1/5] → [5/5]

edit → save → run → observe

Personalise the mission.

Change the values. Keep the variable names.

10:00

01 Give your ship a name.

02 Describe your crew in one sentence.

03 Choose positive whole numbers for oxygen and hull.

04 Choose a galaxy size from 3 to 12.

05 Save and run after each change.

Two common mistakes. Both are fixable.

MISSING QUOTES

```
SHIP_NAME = The Voyager
```

Fix: wrap text in **"quotes"**.

CHANGED THE NAME

```
MY_COOL_SHIP = "The Voyager"
```

Fix: restore **SHIP_NAME**.

Small change? **Test now.**



02

// checkpoint prove it works

Show a neighbour your version.

01

Your ship name appears.

02

The destination count matches.

03

The game finishes without an error.

```
$ uv run game.py
```

Save a snapshot. Share the progress.

status	see the change
add	choose the file
commit	save a snapshot
push	upload it

```
$ git status
$ git add constants.py
$ git commit -m "session 1: personalise my ship"
$ git push
```

DONE WHEN The new commit appears on GitHub.



// reflect say it in your words

Three questions before we land.

01 What is a variable?

02 Which type represents text?

03 Which command uploaded your commit?

a name that stores a value

string

git push

ocean Ai

// session 01 complete

**Your ship has a name.
Your code has a history.**

Next: the game listens to the player.

SESSION 02 → INPUT & CONDITIONALS

